Pearson

T-LEVELS | IfATE Institute for Apprenticeships and Technical Education

T Level Technical Qualification in Digital Production, Design and Development (Level 3)

Wednesday 14 May 2025

Time 3 hours

Paper reference **19538**

Core: Employer Set Project

Task 1:
Planning a project

You do not need any other materials.

Information

- This booklet contains material for the completion of the set task under supervised conditions.
- This booklet is specific to each series and this material must only be issued to students who have been entered to undertake the task in the relevant series.
- This booklet should be kept securely until the start of the supervised assessment period.

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Instructions for students

You **must** complete **ALL** parts of the activity within the assessment.

The task **must** be undertaken at the time and on the date specified by Pearson.

You will be given **3 hours** for producing the outcomes for this task.

Your provider will advise you of when any supervised breaks have been scheduled.

The task **must** be completed under supervised conditions.

You are **not** permitted access to the internet during this task.

You are permitted to use **offline** versions of relevant software to produce evidence for this task.

Your work and any material provided **must** be kept securely at all times.

Set Task Brief

You work for a software development company that has been offered the contract to develop a new digital solution for Bote Parcels.

Your manager has asked you to analyse the potential costs, benefits and risks associated with the project and then produce a plan for the development of the new digital solution.

Your manager intends to use your plan during the next monthly meeting with the client.

Activity

Using the Set Task Information provided, your manager has asked you to produce:

- a Gantt chart to demonstrate how you would organise the development of the new digital solution
- a plan for the resources selected and associated costs
- a rationale that explains your planning approach and justifies the decisions you made.

(19)

Outcomes for submission

Save your Gantt chart as a PDF file **and** as **either**:

- an Excel compatible spreadsheet file
- **or** an MS Project compatible file (as appropriate).

Save your resource and cost plan as a PDF file **and** as **either**:

- an Excel compatible spreadsheet file
- **or** an MS Project compatible file (as appropriate).

Save your rationale as a PDF file.

All files should be saved in your folder for submission.

Use this naming convention:

Task1GanttChart_[Registration number]_[surname]_[first letter of first name]

Task1ResourceCostPlan_[Registration number]_[surname]_[first letter of first name]

Task1Rationale_[Registration number]_[surname]_[first letter of first name]

Set Task Information

Bote Parcels is a growing parcel delivery service. The company collects and delivers parcels across the UK. The company wants to develop a system to manage and support a range of business operations.

After reviewing its current systems, Bote Parcels has decided that it needs to develop a new centralised digital solution.

The aims of the solution are to allow:

- the company to engage with stakeholders more effectively
- employees to manage:
  - customer support requests
  - the collection and delivery of parcels
- management to:
  - access tools for carrying out business data analysis
  - improve the quality of services offered through data analysis.

The requirements of the system are:

- to provide customers with up-to-date information about deliveries, collections and the company's services as a whole
- to allow staff to access information that will support day-to-day activities of the company
- to allow staff to interact with a dashboard of business information
- to provide a user interface with appropriate accessibility features.

Planning information

Bote Parcels estimates that the new solution will lead to an:

- increase in revenue of 7% after the system has been in place for one year
- **additional** increase of 8.5% after two years
- **additional** increase of 4% after three years.

The management team knows that there will be an increase in outgoing costs as a result of the new system.

Bote Parcels and your manager have provided you with information to help you produce your plan.

Company Financial Data	
Description	Amount (£)
Current annual revenue	868 943
Current outgoing	764 624
Forecast for development	Amount (£)
Purchasing of hardware	8 564
Training Bote Parcels' staff to use the new system	975
Software development company base fee	10 750
Server	Option 1: Physical server – 8 750 one-off payment Option 2: Cloud server – 350 per month
Upgrades to company infrastructure	10 500
Your pay (Project Lead)	170 per day
Ongoing costs	Amount (£)
Web hosting and online advertising	255 per month
IT Support Technician	33 000 per year

You have been allocated members of staff by your manager to assist with the project.

Your project development team will consist of these staff members:

- **one** Senior Software Engineer (current employee)
  - Chigozie Ayodele
- **two** Junior Software Engineers (current employees)
  - Venera McReynolds
  - Dorit Sharma
- **one** Hardware and Networking Technician (current employee)
  - Julianna Toft
- **one** Cloud Systems Architect
  - Roland Timberlake (external contractor)

Profiles for the project staff

Name	**Chigozie Ayodele**
Title	Senior Software Engineer
Core skills	• Able to program in multiple languages. • Trained in a range of software testing methods. • Specialises in front-end app and web development.
Previous experience	• Six years' experience in software development. • Worked on implementing small-scale projects within the entertainment sector. • Newly promoted to Team Leader.
Cost	£65 per hour

Name	**Venera McReynolds**
Title	Junior Software Engineer
Core skills	• Able to program in multiple languages including SQL and Python.
Previous experience	• Recently graduated from university with a degree in Software Development. This will be their first job with the development company.
Cost	£25 per hour

Name	**Dorit Sharma**
Title	Junior Software Engineer
Core skills	• Able to program in multiple languages including SQL, HTML/CSS and JavaScript. • Highly skilled in digital graphic design.
Previous experience	• One and a half years' experience as a software engineer for the development company. • Worked as a digital designer for five years before retraining as a software engineer.
Cost	£30 per hour

Name	**Julianna Toft**
Title	Hardware and Networking Technician
Core skills	• Proficient in hardware testing. • Professional qualifications in network infrastructure design and implementation. • Specialist in network design and cybersecurity.
Previous experience	• Five years' experience working as a Hardware and Networking Technician in the manufacturing sector. • One years' experience working for the development company.
Cost	£65 per hour

Name	**Roland Timberlake**
Title	Cloud Systems Architect
Core skills	• Trained in full-stack development. • Proficient in front-end and back-end development, and cloud deployment.
Previous experience	• Five years' experience as a Cloud Systems Architect. • Has worked on large-scale projects developing real-time applications in the financial sector.
Cost	£75 per hour

The table shows the key tasks and the estimated number of work hours it will take to complete each task.

Task	Estimated Hours
Upgrade infrastructure	25
Deploy and configure cloud server	18
Deploy and configure physical server	24
Develop module 1: Customer Relationship Management	90
Develop module 2: Parcel Delivery Management	45
Develop module 3: Vehicle Tracking and Management	80
Develop module 4: User interface	40
Develop module 5: Back-end database	80
Mobile app and Website updates	40
Deploy modules	7 (per module)
Installation of hardware	16
Unit testing	8 (per module)
Integration testing	6 (per module)
Testing of hardware	35
Fixing and regression testing for major fault	7 (per fault)
Fixing and regression testing for minor fault	5 (per fault)
Creating a test plan	2 (per module)
Training Bote Parcels' staff to use the new system	36
User/acceptance testing	40

It is assumed that:

- **three** minor faults will be found with each module
- a maximum of **four** major faults will be found throughout the whole project
- the project will begin on 2nd June 2025
- staff will work eight hours a day, five days a week.

Bote Parcels has requested that the job is completed within 12 weeks.